OCR
Oxford Cambridge and RSA

Tuesday 21 June 2022 – Afternoon

A Level Mathematics B (MEI)

H640/03 Pure Mathematics and Comprehension

Printed Answer Booklet

Time allowed: 2 hours

912202364 2

You must have:
- Question Paper H640/03 (inside this document)
- the Insert (inside this document)
- a scientific or graphical calculator

H640 03

Please write clearly in black ink. **Do not write in the barcodes.**

Centre number | | | | | | Candidate number | | | | |

First name(s) _____

Last name _____

INSTRUCTIONS
- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided in the **Printed Answer Booklet**. If you need extra space use the lined pages at the end of the Printed Answer Booklet. The question numbers must be clearly shown.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.
- Give your final answers to a degree of accuracy that is appropriate to the context.

INFORMATION
- This document has **20** pages.

ADVICE
- Read each question carefully before you start your answer.

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Section A (60 marks)

1	
2(a)	
2(b)(i)	
2(b)(ii)	

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3(a)

3(b)

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4	

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5(a)	

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5(b)

6(a)

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6(b)

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7(a)

7(b)

7(c)

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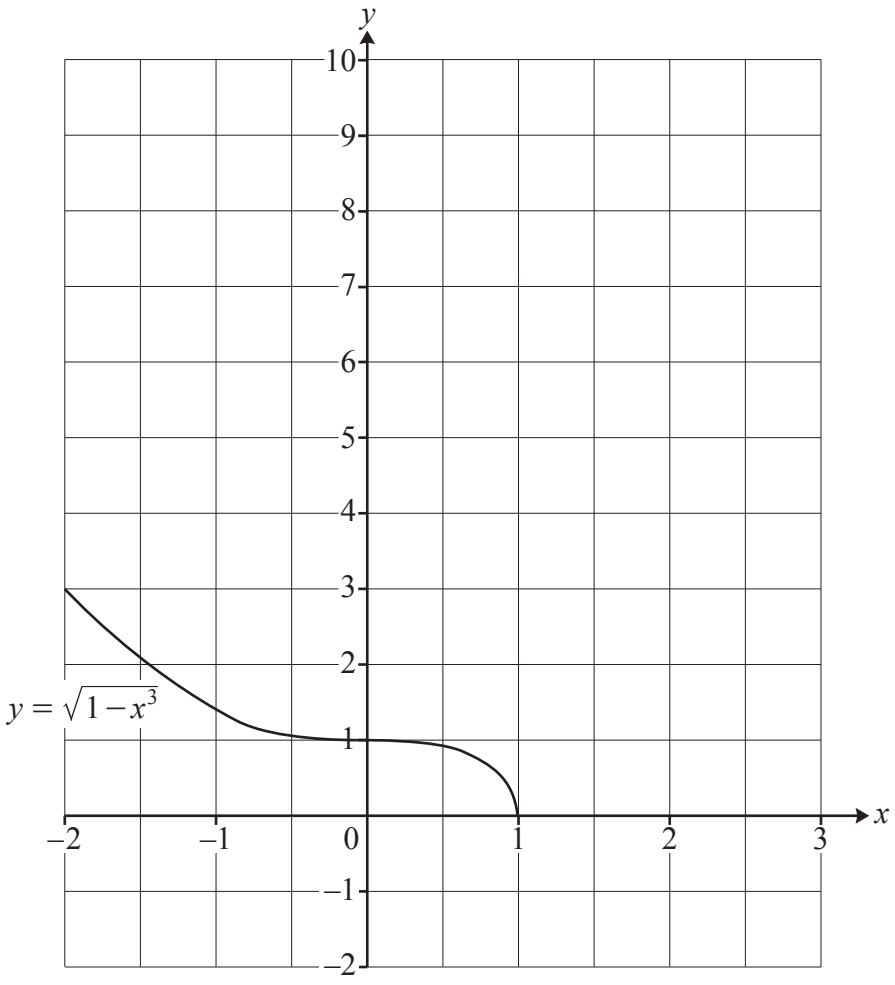
7(d)

7(e)



7(f)

(answer space continued on next page)

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7(f)	**(continued)**

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8(a)

8(b)

Coordinates of A:

Coordinates of B:

Coordinates of C:

Coordinates of D:

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8(c)

8(d)

(answer space continued on next page)

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8(d)	**(continued)**

Section B (15 marks)

The questions in this section refer to the article on the Insert. You should read the article before attempting the questions.

9 Show that $y = x$ has the same gradient as $y = \sin x$ when $x = 0$, as stated in line 5. **[2]**

9	

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10 In this question you must show detailed reasoning.

Fig. C2.2 indicates that the curve $y = \dfrac{4x(\pi - x)}{\pi^2} - \sin x$ has a stationary point near $x = 3$.

- Verify that the x-coordinate of this stationary point is between 2.6 and 2.7.
- Show that this stationary point is a maximum turning point. **[5]**

10

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11 Show that, for the angle 45°, the formula $\sin\theta \approx \dfrac{4\theta(180-\theta)}{40500-\theta(180-\theta)}$ given in line 28 gives the same approximation for the sine of the angle as the formula $\sin x \approx \dfrac{16x(\pi-x)}{5\pi^2-4x(\pi-x)}$ given in line 23. **[3]**

11

12 **(a)** Show that $\cos x = \sin\left(x + \frac{\pi}{2}\right)$. **[2]**

(b) Hence show that $\sin x \approx \frac{16x(\pi - x)}{5\pi^2 - 4x(\pi - x)}$ gives the approximation $\cos x \approx \frac{\pi^2 - 4x^2}{\pi^2 + x^2}$, as stated in line 31. **[3]**

12(a)

12(b)

(answer space continued on next page)

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12(b)	(continued)

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ADDITIONAL ANSWER SPACE

If additional space is required, you should use the following lined page(s). The question number(s) must be clearly shown in the margin(s).

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